



SCHOOL OF COMPUTATION,
INFORMATION AND TECHNOLOGY —
INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Comparative Analysis of Widely Used Zipfian Distribution Implementations

Rodrigo Felix Forno





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**Comparative Analysis of Widely Used
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**Vergleichende Analyse von weit verbreiteten
Zipfian Distribution Implementierungen**

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Submission Date:	Submission date



I confirm that this bachelor's thesis is my own work and I have documented all sources and material used.

Munich, Submission date

Rodrigo Felix Forno

Acknowledgments

Abstract

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1 Introduction

1.1 Section

Acronyms must be added in `main.tex` and are referenced using macros. The first occurrence is automatically replaced with the long version of the acronym, while all subsequent usages use the abbreviation.

E.g. `\ac{TUM}`, `\ac{TUM}` $\Rightarrow$ Technical University of Munich (TUM), TUM

For more details, see the documentation of the `acronym` package¹.

1.1.1 Subsection

See Table 1.1, Figure 1.1, Figure 1.2, Figure 1.3.

Table 1.1: An example for a simple table.

A	B	C	D
1	2	1	2
2	3	2	3

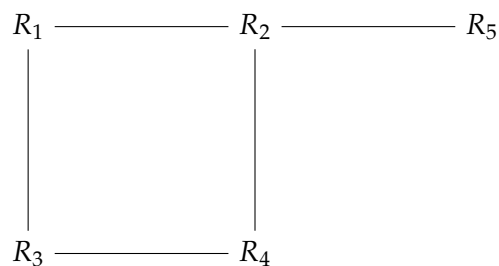


Figure 1.1: An example for a simple drawing.

¹<https://ctan.org/pkg/acronym>

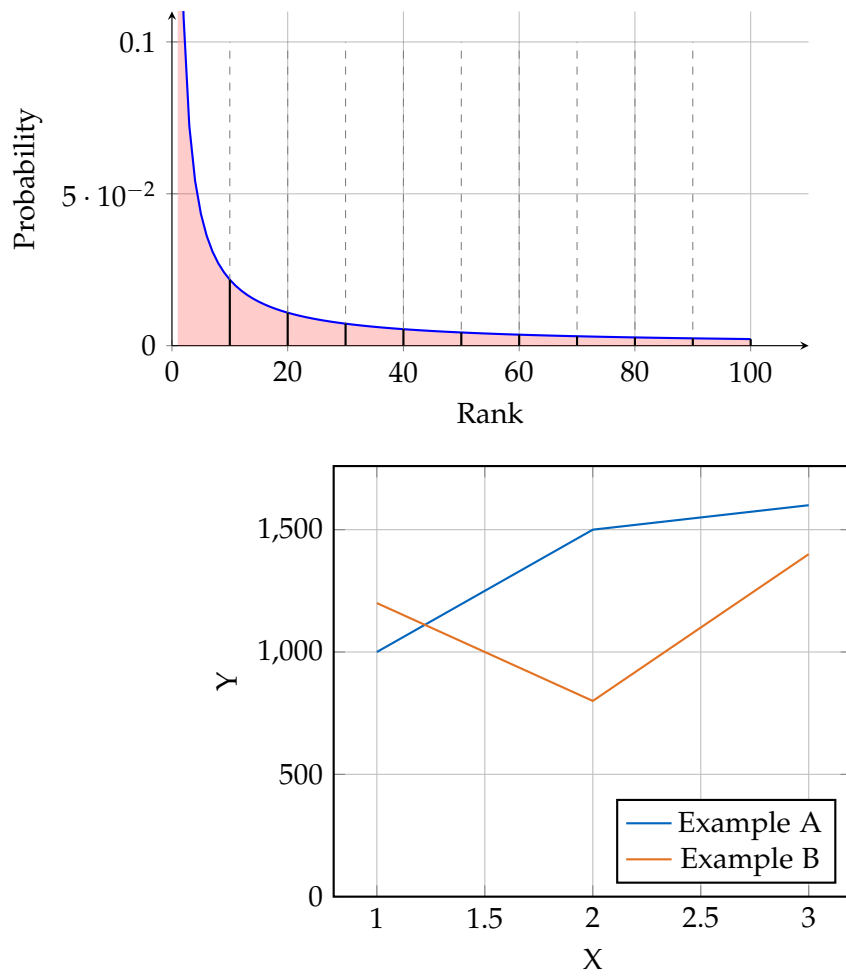


Figure 1.2: An example for a simple plot.

```
SELECT * FROM tbl WHERE tbl.str = "str"
```

Figure 1.3: An example for a source code listing.

2 Approach

The goal of this thesis is to identify the most widely used frameworks for generating zipfian synthetic data in the systems programming and database community. To achieve this, a systematic review of current approaches was conducted with respect to relevancy among high-impact conferences and journals. The following sections describe

1. the **research approach** used to identify the frameworks,
2. the **common frameworks** recognized from the search along with short descriptions
3. the **popularity and impact** of those tools

Afterwards, the **evaluation criteria** used to compare the frameworks are presented and clarified, along with a presentation of the tool used to compile those results.

2.1 Research Approach

To identify relevant frameworks used by the systems programming and database community, first the scope of venues and journals was defined. The list is comprised of:

- Proceedings of the VLDB Endowment (PVLDB)
- ACM Special Interest Group on Management of Data (SIGMOD)
- Conference on Innovative Data Systems Research (CIDR)
- USENIX Symposium on Operating Systems Design and Implementation (OSDI)
- USENIX Symposium on Networked Systems Design and Implementation (NSDI)
- USENIX Conference on File and Storage Technologies (FAST)

These conferences were chosen due to their high selectivity and special relevancy for storage technology research, where there is a prevalence of standardized IO benchmarks. The search was conducted broadly following the steps indicated in [1], as defined below:

1. Word search on keyword search matrix (see table 2.1)
2. Identify the frameworks used in the papers
3. Extended keyword search, this time adding the frameworks identified in the previous step

The first step suggests the use of a keyword matrix. This works by choosing appropriate keywords which refer to the topic of interest. Then for each such keyword, synonyms are used to improve the accuracy and scope of the search. It's defined below as:

	Keyword 1	Keyword 2	Keyword 3
synonyms	zipf zipfian	benchmark workload evaluation empirical dataset	synthetic generated sampled sampler

Table 2.1: Keyword search matrix

The columns are the keywords which are boolean OR'ed, whereas the rows are boolean AND'ed. This means that at least one word of each column should appear in the search.

These queries were then run for the conferences and journals listed before on an unrestricted timescale. The databases consulted are

- IEEE Xplore
- ACM Digital Library
- DBLP
- Google Scholar

On a first phase, the following frameworks were identified among the top 20 results of each conference

Frameworks
Yahoo! Cloud Serving Benchmark
Flexible I/O Tester
Sysbench
OLTP-Bench
DBGen

These frameworks were then searched with the same keyword matrix and their name as a last keyword. This has yielded the following results:

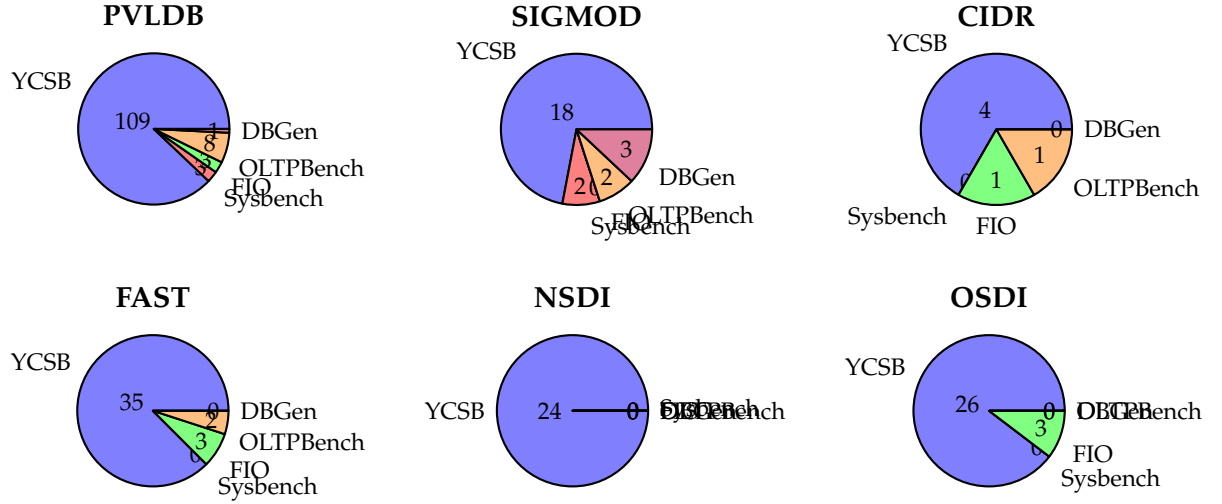


Figure 2.1: Pie charts for each conference (excluding Web Polygraph and pgBench).

Abbreviations

TUM Technical University of Munich

PVLDB Proceedings of the VLDB Endowment

SIGMOD ACM Special Interest Group on Management of Data

CIDR Conference on Innovative Data Systems Research

OSDI USENIX Symposium on Operating Systems Design and Implementation

NSDI USENIX Symposium on Networked Systems Design and Implementation

FAST USENIX Conference on File and Storage Technologies

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Bibliography

- [1] W. M. Bramer, G. B. de Jonge, M. L. Rethlefsen, F. Mast, and J. Kleijnen, "A systematic approach to searching: An efficient and complete method to develop literature searches," *Journal of the Medical Library Association : JMLA*, vol. 106, no. 4, pp. 531–541, Oct. 2018, issn: 1536-5050. doi: 10 . 5195 / jmla . 2018 . 283. pmid: 30271302.